Generator-induced invariants in lattice public keys: a pre-standardization rank diagnostic

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Abstract

A public-key construction that builds its public object by applying a fixed family of linear maps to random seeds—to compress the key, to inherit a symmetry, or to support a reduction—produces an object that is the image of a single fixed *design matrix*. Every invariant of that matrix, its rank first of all, is then imposed on the public object whatever the seeds. When the rank ρ of the design matrix is below the ambient dimension, the public key is distinguishable from uniform by exact linear algebra, and recovering the noise reduces to an LWE instance of dimension at most ρ. The argument takes one line—the rank of a product is at most the rank of either factor—yet the weakness is easy to miss: a rank check on a realized instance of dimension n can certify only $\rho \geq n$, so a check run below the ceiling returns a clean result on a broken scheme. We make this concrete with a reproducible case study, a withdrawn sedenion Module-LWE proposal whose public matrix has rank at most 76 at every module rank and whose own structural check, run at module rank 4, could not see the ceiling. We package the lesson as an exact and cheap diagnostic—compute the rank of the design matrix over the working field, then confirm it on an instance larger than the ceiling against a uniform control—with a dependency-free implementation that reproduces the case study, and we place the mechanism among published breaks in which structure introduced for a benefit became the attack surface. The diagnostic is necessary, not sufficient, and costs less than the lattice estimate it should precede.

Epistemic conventions. As in the author's other work, claims are tiered. Every proposition and corollary below is T1: proven from the stated definitions, or verified by exact computation. Section 5 classifies published breaks; that classification is interpretive, and where it reaches beyond Proposition 1 it says so.

1 Introduction

Uniformly random public keys are large, and designers routinely replace them with a *generated* object that is cheaper to store or transmit: the cyclic and negacyclic structure of Ring- and Module-LWE, the quasi-cyclic and dyadic codes of compact code-based schemes, group-action images, and orbit constructions of many kinds. The generator is always introduced for a reason: a smaller key, a symmetry the designer believes is benign or even helpful, or a structural argument that the construction reduces to a well-studied problem.

The thesis of this note is that such a generator is a fixed linear gadget, and that random seeds fed through a fixed linear gadget cannot produce anything outside its image. If the gadget carries an invariant—a rank ceiling, a fixed subspace, a common kernel—the public object inherits it however much entropy the seeds carry. When the invariant is a rank ceiling below the ambient dimension, the consequences are immediate: the public key is distinguishable from uniform, and recovery drops from the ambient dimension to at most the rank of the design matrix.

None of the linear algebra is deep, and that is the point. The mechanism is elementary, and the diagnostic that catches it takes seconds of exact arithmetic. Yet the weakness is easy to ship, because a rank check on a realized instance certifies only that the ceiling is at least the size of that instance. Below the ceiling, a generated object can be full rank and indistinguishable from a uniform control; a check run there—the diligent thing to do—returns a clean result. A scheme can therefore pass its own structural test and still carry a fatal ceiling, if the test was run on too small an instance.

Contributions.

1. A general statement (Section 2): a seed-generated public object factors as $M = R \cdot D$ with D a fixed design matrix, so $\operatorname{rank} M \leq \operatorname{rank} D$ for every instance size and every draw of the seeds. The same factorization gives a kernel shared by every public key (a distinguisher) and confines the LWE error to a subspace of dimension at most $\operatorname{rank} D$ (a reduced recovery problem).

2. A visibility principle (Proposition 2): a full-rank check at ambient dimension n certifies only $\operatorname{rank} D \geq n$, and a deficiency is guaranteed to show only once $n > \operatorname{rank} D$. Computing $\operatorname{rank} D$ from the design certifies every size at once.

3. A reproducible case study (Section 3): a withdrawn sedenion Module-LWE proposal in which exactly this mechanism gives a ceiling of 76 at every module rank, with an account of why the scheme's own check, run at module rank 4, could not see it.

4. An exact, cheap diagnostic (Section 4) with a dependency-free implementation that reproduces the case study as a regression test.

5. A short classification of published breaks of the same shape (Section 5), with an account (Section 6) of what the theorem does and does not cover.

Scope. This is a defensive, methodological note. The case study is the author's own withdrawn proposal, and the NIST standards [16] appear only as correctly designed controls. No attack on any deployed or standardized system is given or implied.

2 The mechanism

2.1 Setup

Let $\mathbb{F}$ be a field. A *seed-generated public object* is built as follows. The scheme fixes a finite family of $\mathbb{F}$-linear maps $T_0, \ldots, T_{r-1} \colon \mathbb{F}^d \to \mathbb{F}^d$ (the *generator*), draws independent random seeds, applies the generator to the seeds, and combines the results $\mathbb{F}$-linearly. Write M for the matrix of the $\mathbb{F}$-linear map the scheme exposes—for an LWE-type key, the map $s \mapsto As$. Because the assembly is linear in the seeds and linear in the fixed generator images, M is bilinear in the two and factors as

$$M \;=\; R \cdot D,$$

where $R \in \mathbb{F}^{a \times N}$ carries all of the seed randomness (its entries are linear in the seed coordinates) and $D \in \mathbb{F}^{N \times b}$ is *fixed*: its entries are determined by the generator and the combination rule and do not depend on the seeds. We call D the *design matrix*.

2.2 The rank ceiling

Proposition 1 (rank ceiling)**.** *For every instance and every draw of the seeds,* $\operatorname{rank} M \leq \operatorname{rank} D$.

Proof. $M = RD$ and $\operatorname{row}(RD) \subseteq \operatorname{row}(D)$, so $\operatorname{rank} M \leq \operatorname{rank} D$. □

The content lies entirely in the fact that D is fixed. Seed entropy moves M within the row space of D; it cannot raise the rank above $\operatorname{rank} D$. If $\operatorname{rank} D$ is below the ambient dimension, every public key the scheme can emit is rank-deficient, with probability one.

Corollary 1 (distinguisher)**.** *Let* $\rho = \operatorname{rank} D < b$. *Then* $\ker D \subseteq \ker M$ *for every public key, and* $\dim \ker D = b - \rho$. *In particular, for any fixed nonzero* $x \in \ker D$*—computable from the design alone—every public key satisfies* $Mx = 0$, *while a uniform* $M \in \mathbb{F}_q^{a \times b}$ *does so with probability* q^{-a}.

Proof. If $Dx = 0$ then $Mx = R(Dx) = 0$; the dimension count is rank–nullity. For uniform M, each of the a rows is orthogonal to $x \neq 0$ independently with probability $1/q$. □

Corollary 2 (recovery)**.** *Write the public key in LWE form* $t = Ms + e$ *with* $M \in \mathbb{F}_q^{a \times b}$, *and let* $V = \operatorname{col}(M)$ *and* $\rho_M = \dim V \leq \operatorname{rank} D$. *Then* $t - e \in V$, *so* e *lies in the known affine subspace* $t + V$ *of dimension* ρ_M. *Choosing* ρ_M *coordinates* I *on which a basis of* V *is invertible expresses the remaining* $a - \rho_M$ *coordinates of* e *as fixed affine functions of* e_I*: an LWE instance with secret* e_I *of dimension* ρ_M *and* $a - \rho_M$ *samples.*

Proof. Every $v \in V$ is determined by v_I, with $v_{\bar{I}} = Cv_I$ for a matrix C computable from M. Applied to $v = t - e$ this gives $e_{\bar{I}} = Ce_I + (t_{\bar{I}} - Ct_I)$, in which e_I and $e_{\bar{I}}$ are short. □

Remark 1. Rank is the cheapest invariant to read off, but not the only one: any fixed subspace that D preserves or annihilates transfers to M by the same argument and gives a projection that shrinks the secret. The rank ceiling is the easiest to test and, when present, the most damaging.

2.3 Why the ceiling hides

The design matrix of an instance is assembled from the generator images that instance uses. Suppose the generator contributes c independent images in total; for a cyclic generator of order r acting on a d-dimensional alphabet, $c \leq rd$, and the column space of the design stops growing once an instance uses all r powers. Let $\rho = \operatorname{rank} D$ for the full design, let M_k be the realized public map at instance size k, and let $n(k)$ be the smaller dimension of M_k, which is the rank a uniform matrix of that shape has with high probability.

Proposition 2 (visibility)**.**

(i) If $\operatorname{rank} M_k = n(k)$, *then* $\rho \geq n(k)$, *and Proposition 1 excludes no larger value of* ρ*: a full-rank result at size* k *is consistent with every ceiling* $\rho \geq n(k)$, *including ceilings far below the ambient dimension* $n(k')$ *of a larger deployed size* k'.

(ii) If $n(k) > \rho$, *then* $\operatorname{rank} M_k \leq \rho < n(k)$*: the deficiency is visible.*

Proof. Proposition 1 and the shape of M_k give $\operatorname{rank} M_k \leq \min(n(k), \rho)$. If $\operatorname{rank} M_k = n(k)$, this forces $n(k) \leq \rho$ and constrains ρ no further. Table 1 exhibits a design with $\rho = 76$ whose realizations have full rank at every $k \leq 4$ ($n(k) \leq 64$), although the deployed size has $n(k') = 1024$. Part (ii) is immediate. □

Remark 2. In the case study the realized rank is exactly $\min(n(k), \rho)$ at every size tested (Table 1): full rank up to the ceiling, and pinned at ρ once the generator's images repeat ($k \geq 7$, since $\sigma^7 = \mathrm{id}$). The deficiency therefore first appears at the smallest k with $n(k) > \rho$, here $k = 5$, and a check at $k = 4$ is precisely the uninformative case (i). This "full rank up to the ceiling" behaviour is a property of the case-study design, established by exact computation. It is not a theorem about all designs; a design that falls short before its ceiling simply shows its deficiency earlier.

Two practical rules follow. An empirical rank check certifies only the size it is run at, so a check below the deployed size certifies nothing about the deployed key. And computing $\rho = \operatorname{rank} D$ from the design certifies every size at once: it is cheaper and stronger than any empirical check, which then serves only to confirm that the factorization was done correctly.

3 Case study: a sedenion Module-LWE proposal

We instantiate the mechanism on a concrete, withdrawn scheme. The numerical claims in this section are those of the negative-result note that broke it [11], to which we refer for the computations; the aim here is to exhibit the general mechanism, not to re-derive the break. The scheme and the break are the author's own.

3.1 The construction and its intent

The scheme, SLWE [10], is a Module-LWE key encapsulation mechanism [17, 14] over the 16-dimensional sedenion algebra $\mathbb{S}_q = \mathbb{S} \otimes \mathbb{F}_q$, the fourth Cayley–Dickson algebra. Its public matrix $A \in \mathbb{S}_q^{k \times k}$ is built from orbits of an order-7 coordinate permutation σ—a "Singer cycle" inherited from the symmetry of the Fano plane—as $A_{ij} = \sigma^j(\mathrm{seed}_i)$, with independent uniform seeds $\mathrm{seed}_i \in \mathbb{S}_q$. The orbit structure was deliberate: it compresses the key (each row comes from one seed) and was meant to preserve the algebra's zero-divisor interface. Non-associativity was argued to remove an ideal-lattice attack class and so to reduce security to that of a generic module lattice. Two parameter sets were published with a claim of NIST Category 5: $n = 512$ ($k = 32$) with $q \approx 2^{32}$, and $k = 64$ ($n = 1024$) with $q = 911$.

3.2 The factorization and the ceiling

Let $M \in \mathbb{F}_q^{16k \times 16k}$ be the matrix of $s \mapsto As$, so that $M_{16i+u,\,16j+c}$ is coefficient u of $\sigma^j(\mathrm{seed}_i)\, e_c$. Because σ is linear and the sedenion product is bilinear, M factors exactly as in Section 2:

$$M = (S \otimes I_{16})\, N_k,$$

where $S \in \mathbb{F}_q^{k \times 16}$ holds the seeds (this is R) and N_k is assembled from the columns of a fixed integer matrix $N \in \mathbb{Z}^{256 \times 112}$ whose entry $N_{16m+u,\,16j+c}$ is coefficient u of $\sigma^j(e_m)\, e_c$ (this is D). Since $\sigma^7 = \mathrm{id}$, the column blocks of N_k repeat with period 7, so $\mathrm{col}(N_k) \subseteq \mathrm{col}(N)$, with equality once $k \geq 7$.

Exact computation gives $\operatorname{rank} N = 76$ over $\mathbb{Q}$ and modulo 911 and $q = 4\,294\,977\,961$ [11]; the accompanying diagnostic re-derives it modulo 911, 2731, $2^{31} - 1$ and q. By Proposition 1, $\operatorname{rank} A \leq 76$ at every module rank, against $16k$ for a uniform matrix. Table 1 lists the exact ranks.

By Corollary 2, recovering the noise is an LWE instance of dimension 76 with $16k - 76$ samples. The lattice estimator [2, 3] places it at $2^{38.6}$ operations at the first parameter set (436 samples) and at $2^{40.4}$ at the second (948 samples), with the block size at the floor of its model, so that reduction at the scale of LLL suffices [11]. Distinguishing the public key from uniform is cheaper still: by Corollary 1, it takes Gaussian elimination.

k	1	2	3	4	5	6	7	8	16	32	64
$n(k) = 16k$	16	32	48	64	80	96	112	128	256	512	1024
rank M_k (Singer orbit)	16	32	48	64	**72**	**74**	**76**	**76**	**76**	**76**	**76**
rank M_k (uniform)	16	32	48	64	80	96	112	128	256	512	1024

Table 1: Exact rank of the flattened public map at module rank k, Singer-orbit versus uniform matrix, over $\mathbb{F}_{911}$ and over $\mathbb{F}_q$ with $q = 4\,294\,977\,961$ (identical at both primes), as reported in [11]; the columns $k \leq 8$ are regenerated by the self-test of the accompanying diagnostic (Section 4). Bold entries fall short of $n(k)$. The columns $k \leq 4$ are identical for the two matrices: a check there cannot see the ceiling (Proposition 2(i)).

3.3 Why the scheme's own check missed it

The proposal included a structural self-test: a five-test Gram–Schmidt profile of the lattice, looking for exactly the kind of anomaly at issue. It reported no rank deficiency, no anomalous short vectors and no invariant sublattice relative to a uniform matrix [10]. It was run on a 64×64 proxy at module rank $k = 4$, and the proposal itself labelled it a toy-scale result that "does not establish anything at production dimension".

Proposition 2 makes that caveat exact. At $k = 4$ the public map has $n(4) = 64$ columns, below the ceiling of 76, so by part (i) a full-rank result certified only $\rho \geq 64$. The deployed matrix at $k = 64$ has $n = 1024$; every value $64 \leq \rho < 1024$ was left open, and the truth was $\rho = 76$. The check returned a clean result not because the scheme was sound but because it was run below the ceiling. Table 1 shows the trap at a glance: the columns $k \leq 4$ are identical for the Singer-orbit and the uniform matrix, and they separate only from $k = 5$.

3.4 Non-associativity was beside the point

The scheme's security narrative rested on the non-associativity of the sedenions and on the σ-symmetry of A. Neither bears on the break. Decryption uses the bilinear conjugate-norm pairing, so every map the attacker needs is $\mathbb{F}_q$-linear, and the argument of Section 2 uses only linearity. The permutation σ is not an automorphism of the sedenion product but a relabelling of coordinates, so the "symmetry" of A is not an algebraic symmetry at all [11]. The lesson travels: advertised algebraic richness can be orthogonal to the property that carries security, and a security argument phrased in terms of the rich structure may be answering the wrong question.

3.5 The repair that was not one

Once the deficiency was found, the natural repair was to replace the Singer-orbit matrix with a uniform one, which removes the ceiling. It does not rescue the claim: at the first parameter set the modulus-to-noise ratio alone puts the cheapest attack near 2^{51}, and at the second near 2^{167}, about 95 bits below the 2^{262} the same estimator gives ML-KEM-1024 [11]. Clearing the structural flag one has found does not clear a parameter problem one has not. The rank test of Section 4 is necessary, never sufficient.

4 The diagnostic

The mechanism gives a checklist that a designer or evaluator can run before trusting a compressed or structured public key, at a cost far below the lattice estimate it should precede.

1. **Factor.** Write the construction as $M = R \cdot D$, separating the seed randomness R from the fixed generator gadget D. If no such factorization exists because the public object is

uniform, there is nothing to check here.

2. **Build the full design.** Include every distinct generator image, so that D covers all instance sizes (for a cyclic generator of order r, all r powers).

3. **Compute the ceiling exactly.** Compute $\rho = \operatorname{rank} D$ over the working field $\mathbb{F}_q$ with exact arithmetic, not floating point, and over a second characteristic to confirm that the value is not an artifact of one prime. This bounds $\operatorname{rank} M$ at every instance size at once (Proposition 1).

4. **Confirm on a large enough instance.** Realize M at a size with $n(k) > \rho$ and compare its rank with a uniform matrix of the same shape over several seed sets. A check below that size is uninformative (Proposition 2); a realized rank above ρ contradicts Proposition 1 and means the factorization is wrong.

5. **Flag.** Report a ceiling if ρ is below the ambient dimension at the deployed size, and a projection weakness if D preserves or annihilates a nontrivial fixed subspace. Either gives a distinguisher (Corollary 1) and a reduced recovery problem (Corollary 2).

The test is exact linear algebra on a matrix the size of the design: it takes seconds, needs no lattice reduction or estimator, and costs less than the Core-SVP estimate a candidate already runs. It belongs *before* that estimate, because a positive result makes the estimate moot.

Where to look. The factorization of step 1 exists whenever the public object is a generator applied to seeds: cyclic or orbit-generated matrices, group-action public keys, structured-code substitutions, and any object described as compressed, symmetric, or inheriting an algebraic structure. A uniform key, for which no such factorization exists, is the sound design this test compares against.

Implementation. A standalone implementation in pure Python, with no dependencies, accompanies this note (`design_matrix_rank.py`; Section 7). A user describes a construction by implementing four methods: `design_matrix()`, `realize()`, `ambient_rank()` and `random_seeds()`. The tool then runs steps 3 and 4 and the rank part of step 5, and prints a verdict. It ships with the case study of Section 3 (verdict *fail*: ceiling 76, the same over four characteristics) and a uniform control (verdict *pass*). A `--self-test` mode reproduces the columns $k \leq 8$ of Table 1 over three seed sets and $\operatorname{rank} N = 76$ over four characteristics, and exits with an error on any mismatch, so it doubles as a regression test pinned to the published result. The empirical curve it prints marks every size below the ceiling as one at which the collapse is invisible: Section 3.3 in executable form.

5 The pattern recurs

The case study is one instance of a recurring shape: *a structural feature introduced for a benefit carries an invariant, or an auxiliary datum, that an attacker reads off.* Only the first item below is an instance of Proposition 1. The others are published breaks, cited as established results; what we add is the classification—that each turns a structure introduced for a benefit into the lever—which is interpretive, not a theorem. All are breaks of proposed or withdrawn constructions.

- **A rank ceiling from a generator.** The case study of Section 3. Benefit sought: a compressed key and an inherited symmetry. Lever: the rank of the design matrix. This is the linear-image case that Proposition 1 governs directly.

- **Structured codes in McEliece.** Random binary Goppa codes give a sound McEliece instance, but substituting a structured code to shrink the key has failed repeatedly: generalized Reed–Solomon codes [19], compact quasi-cyclic and dyadic variants [9], and algebraic-geometry codes [6], each broken by a structural attack that recovers the hidden code from the public generator matrix. Structure added for size became the lever.

- **Overstretched NTRU.** With a large modulus at fixed noise, a dense sublattice appears, which subfield and sublattice attacks exploit [1, 13]; the onset was later located as the "fatigue point" [8]. Here the structure comes from a parameter choice made for functionality rather than from an explicit generator, but the lever is again a fixed low-dimensional structure in the public object.

- **Principal ideals and short generators.** Compact key constructions over principal ideals, such as Soliloquy [4], admit short-generator recovery [7]: the ideal structure chosen for compactness is the route to the key.

- **SIDH and SIKE: auxiliary data for functionality (an analogy).** The images $\phi(P), \phi(Q)$ of torsion points are published so that the key exchange can complete. The 2022 attacks, all published at EUROCRYPT 2023 [5, 15, 18], use Kani's theorem [12] to embed the secret isogeny in a higher-dimensional abelian variety and turn exactly that auxiliary data into key recovery, in polynomial time in full generality [18]. This is not a rank ceiling and not an instance of Proposition 1; it marks the boundary of the linear-image argument while sharing its moral: the structure published to make the scheme work was the attack surface.

The question this classification recommends asking of any structured proposal is simple: *what did the structure I added give the attacker?* For the linear-image cases, Section 4 answers it mechanically and cheaply; for the others it is a design discipline rather than a single test.

6 Limitations

- **Linear-image generators only.** Proposition 1 is exact and general for public objects bilinear in the seeds and a fixed generator. Nonlinear generators, and the SIDH case of Section 5, lie outside it; there the comparison is an analogy, and Section 5 says so.

- **Necessary, not sufficient.** A full-rank design matrix with no invariant subspace shows only that this class of weakness is absent. Section 3.5 is the cautionary case: with the structure removed, the scheme still failed on its modulus-to-noise ratio, which this test does not judge.

- **Structure is not the enemy.** Module-LWE and binary-Goppa McEliece are structured and sound. The claim is narrow and testable: generator-induced invariants must be computed, over the working field and for the full design, before a structured public object is trusted.

- **Dimension, not cost.** Corollary 2 gives the dimension to which recovery reduces; the concrete cost is a lattice-estimator question that depends, as Section 3.5 shows, on parameters the rank test does not see.

7 Reproducibility

The diagnostic is a single file in the public repository `gifgaf0/gifgaf0.github.io`, and a copy accompanies this note on Zenodo:

file	`generator_induced_invariants/diagnostics/design_matrix_rank.py`
SHA-256	`bb22d680f7b82e3d69e722923654a87ea4c558cbe9f932ea4c1a0b9a0f401b94`
self-test	`python3 design_matrix_rank.py --self-test`

It runs under Python 3.8 or later with no other dependencies. The self-test reproduces the columns $k \leq 8$ of Table 1 and rank $N = 76$ modulo 911, 2731, $2^{31} - 1$ and $q = 4\,294\,977\,961$, by exact Gaussian elimination over $\mathbb{F}_p$. The full break of the case-study scheme—both independent rank implementations, one written blind from a textual specification, the exact ranks computed with python-flint, and the lattice-estimator runs—is archived with [11].

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